

LegalEase

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1.1 User classes and Characteristics

Chapter 1

Introduction

This document describes the requirements and design of **LegalEase AI**, a web-based platform developed to assist lawyers, clients, and law students in managing legal information and activities more efficiently. The system integrates modern web technologies with artificial intelligence to provide features such as case management, AI-based legal assistance, document analysis, and a legal practice simulator. The goal of the platform is to simplify legal workflows, improve access to legal information, and support learning and decision-making in the legal domain.

The purpose of this document is to present the functional and non-functional requirements of the system and to explain how different modules of the application will work together. It also provides an overview of the system architecture and the expected behavior of the platform.

This document is intended for multiple readers. **Developers** will use it to understand the system structure and implement the application modules. **Project supervisors and project managers** may review it to evaluate the project progress and scope. **Testers** can use it to design test cases for validating the system functionality. In addition, **users such as lawyers, clients, and students** can refer to this document to understand the purpose and capabilities of the platform

1.1 Problem Statement

The legal sector still relies heavily on manual processes for managing cases, legal documents, and communication between lawyers and clients. Many lawyers handle multiple cases at the same time, which makes it difficult to track important deadlines, maintain documents, and monitor the progress of each case efficiently. Important information such as hearing dates, case notes, and client communications are often scattered across different sources or managed manually, which increases the chances of mistakes or missed deadlines. Clients also face difficulties in getting regular updates about their cases and usually depend on direct communication with their lawyers for even small updates. In addition, individuals who are not familiar with legal procedures often struggle to understand legal terminology and processes. Law students also lack practical platforms where they can practice legal scenarios and improve their understanding of real-world legal cases.

The purpose of developing **LegalEase AI** is to address these issues by providing a centralized digital platform that simplifies case management and legal interaction. The system will help

lawyers organize their cases, manage important deadlines, store and analyze legal documents, and collaborate with clients more effectively. Clients will be able to access updates about their cases through a secure online platform instead of relying only on manual communication. The system will also include an AI-based legal assistant that can provide basic legal guidance and answer common legal queries. Furthermore, a practice simulator will allow students to explore legal scenarios and improve their understanding of legal procedures. By integrating these features into a single platform, LegalEase AI aims to make legal information management more organized, accessible, and efficient.

1.2 Scope

The scope of the **LegalEase AI** project is to develop a web-based platform that supports legal professionals, clients, and law students by providing tools for managing legal information and interacting with AI-assisted services. The system focuses on simplifying the process of managing legal cases, storing documents, tracking deadlines, and improving communication between lawyers and clients. Lawyers will be able to create and manage cases, upload and organize case-related documents, set hearing dates or deadlines, and collaborate with clients through the platform. Clients will be able to view updates related to their cases, access shared documents, and stay informed about important case activities. The system will also include an AI-based legal chat assistant that can answer legal questions and provide basic legal guidance using pre-trained models and legal datasets.

Another important part of the project is to support law students through a **legal practice simulator**, where they can explore different legal scenarios and understand how cases may be handled in real situations. The system will also include features for document analysis and legal research to assist users in reviewing legal content more efficiently. The project mainly focuses on providing an organized digital environment for managing legal information rather than replacing professional legal advice. Overall, the scope of the system is limited to building a functional web platform that integrates case management, AI-based assistance, document handling, and learning tools in a structured and user-friendly way.

1.3 Modules

1.3.1 Case Management Module

This module allows lawyers to create and manage legal cases within the system. It provides tools for organizing case details, storing documents, tracking important deadlines, and linking clients with their respective cases.

Features

1. Create and manage case records
2. Upload and organize case-related documents
3. Track hearing dates and important deadlines
4. Add clients as collaborators to cases

1.3.2 AI Legal Chat Assistant

This module provides an AI-powered chat interface where users can ask legal questions and receive informative responses. The system uses pre-trained natural language processing models to understand queries and provide helpful legal guidance.

Features

1. Ask legal questions through chat interface
2. Receive AI-generated legal explanations
3. Support queries in both Urdu and English

1.3.3 Document Analysis Module

This module helps users analyze legal documents by extracting text and identifying key clauses or summaries. It assists lawyers in quickly reviewing legal content and understanding important parts of documents.

Features

1. Upload and analyze legal documents
2. Extract important information from documents
3. Generate summaries of legal text

1.3.4 Legal Research Module

This module allows users to search and explore legal information such as case references and legal concepts. It helps lawyers and students access relevant legal knowledge more efficiently.

Features

1. Search legal information and references
2. Explore related legal concepts and cases

1.3.5 Practice Simulator Module

The practice simulator provides a learning environment for law students to explore legal scenarios and practice decision-making. It simulates different case situations and allows students to interact with the system for learning purposes.

Features

1. Simulate legal case scenarios
2. Allow students to analyze and respond to cases

1.3.6 Contract Drafting And Compliance Module

Assist in creating, reviewing, and verifying contracts.

Features:

1. Drafting basic legal contracts
2. Suggesting clauses and reviewing versions.
3. Check compliance with legal templates and rules.
4. Version comparison and review support.

1.3.7 OCR And Scanned Document Module

Convert scanned or image-based legal documents into editable text

Features:

1. OCR processing for PDFs or images
2. Text extraction from scanned FIRs, contracts, and judgments.
3. Integration with document analysis and AI modules.

1.3.8 Security, Role Management And Notifications Module

Ensure data protection, controlled access, and system communication.

Features:

1. Role-based access control for lawyers, clients, and students.
2. Encrypted storage for sensitive documents and legal information.
3. Activity and audit logging for accountability.
4. Notifications for deadlines, updates, and document changes.

1.3.9 Unified Dashboard and User interface Module

Provide role-based dashboards for Lawyers, Clients, and Students/Interns through a single structured interface.

Features:

1. Separate dashboards based on user role (Lawyer, Client, Student/Intern).
2. Overview of recent activities, notifications, and pending tasks.
3. Quick access to cases, documents, and AI tools.
4. Personalized workspace based on permissions.
5. Responsive and user-friendly interface.

1.4 User Classes and Characteristics

The LegalEase AI platform is designed to support different types of users, each with specific roles and system interactions.

User Class	Description
Lawyer	Lawyers are the primary users of the system. They will use the platform to create and manage legal cases, upload documents, track important deadlines, and

	collaborate with their clients. Lawyers are expected to have basic knowledge of computer systems and web applications.
Client	Clients are individuals who have legal cases handled by lawyers using the system. They can log in to view updates about their cases, access shared documents, and stay informed about deadlines or progress related to their legal matters.
Student	Law students will use the platform mainly for learning purposes. They can interact with the practice simulator to explore legal scenarios and improve their understanding of legal procedures. Students may also use the AI assistant to learn about legal concepts.

Chapter 2

Project Requirements

2.1 Use Case Diagram:

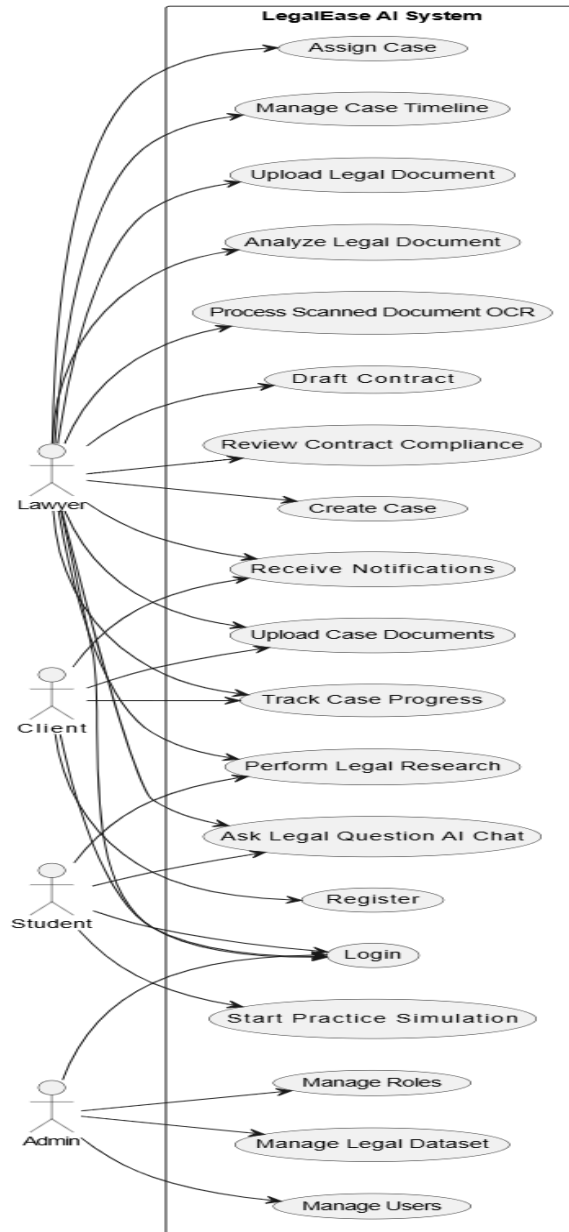


Figure: Use Case Diagram

2.2 Functional Requirements

2.2.1 Case Management Module

This module manages the lifecycle of legal cases.

- **FR-CM-01:** The system shall allow lawyers to create a new legal case.
- **FR-CM-02:** The system shall allow lawyers to assign cases to themselves or other authorized users.
- **FR-CM-03:** The system shall allow users to upload case-related documents.
- **FR-CM-04:** The system shall store and organize documents associated with each case.
- **FR-CM-05:** The system shall allow lawyers to track the progress of legal cases.
- **FR-CM-06:** The system shall allow lawyers to record important events such as hearings or deadlines.
- **FR-CM-07:** The system shall provide a visual timeline of case activities.
- **FR-CM-08:** The system shall allow clients to view the progress of their assigned cases.

2.2.2 Legal Document Processing Module

This module handles document analysis.

- **FR-DOC-01:** The system shall allow users to upload legal documents in formats such as PDF or text.
- **FR-DOC-02:** The system shall analyze uploaded legal documents using
- **FR-DOC-03:** The system shall generate summaries of legal documents.
- **FR-DOC-04:** The system shall identify important clauses within legal documents.
- **FR-DOC-05:** The system shall highlight potential risks or missing clauses.

2.2.3 AI Legal Research Module

This module assists users in finding relevant legal information.

- **FR-RES-01:** The system shall allow users to search for legal cases
- **FR-RES-02:** The system shall retrieve relevant legal cases from the legal dataset.
- **FR-RES-03:** The system shall provide semantic search results based on query similarity.
- **FR-RES-04:** The system shall allow users to filter results based on case type or legal domain.
- **FR-RES-05:** The system shall suggest similar legal cases.

2.2.4 AI Legal Chat Assistant Module

This module provides AI-based legal guidance.

- **FR-AI-01:** The system shall allow users to ask legal questions through a chat interface.
- **FR-AI-02:** The system shall generate responses using an AI language model.
- **FR-AI-03:** The system shall retrieve relevant legal documents to support AI responses.
- **FR-AI-04:** The system shall present explanations of legal concepts in simplified language.

2.2.5 Contract Drafting & Compliance Module

This module assists in creating and reviewing contracts.

- **FR-CON-01:** The system shall allow users to create draft contracts.
- **FR-CON-02:** The system shall suggest relevant clauses during contract drafting.
- **FR-CON-03:** The system shall check contracts for compliance with predefined templates.
- **FR-CON-04:** The system shall allow users to review and modify contract drafts.

2.2.6 OCR & Scanned Document Module

This module processes scanned legal documents.

- **FR-OCR-01:** The system shall allow users to upload scanned legal documents.
- **FR-OCR-02:** The system shall extract text from scanned images or PDFs using OCR.
- **FR-OCR-03:** The system shall convert extracted text into editable digital format.
- **FR-OCR-04:** The system shall store extracted text for further analysis.

2.2.7 Practice Simulator Module

This module provides legal training scenarios.

- **FR-PS-01:** The system shall allow students or lawyers to start a simulated legal practice session.
- **FR-PS-02:** The system shall simulate interactions with a virtual client or judge.
- **FR-PS-03:** The system shall provide scenario-based legal practice exercises.
- **FR-PS-04:** The system shall provide feedback on user responses.

2.2.8 Notification Module

This module keeps users informed of system events.

- **FR-NOT-01:** The system shall notify users of case updates.
- **FR-NOT-02:** The system shall notify users of upcoming deadlines.
- **FR-NOT-03:** The system shall notify users when new documents are uploaded to a case.

2.3 Non-Functional Requirements

2.3.1 Reliability

Reliability refers to the ability of the system to perform its required functions under stated conditions for a specified period.

- **REL-1:** The LegalEase system shall maintain **99% system uptime** excluding scheduled maintenance.
- **REL-2:** The system shall ensure that no case data or user information is lost during unexpected failures by implementing **automatic database backups every 24 hours**.
- **REL-3:** The system shall recover from system crashes within **5 minutes** without loss of saved case records or documents.
- **REL-4:** In the event of AI service failure, the system shall notify the user and allow continued access to other modules such as **case management and document storage**.
- **REL-5:** All user transactions such as **case creation, updates, and document uploads** shall maintain data integrity even during network interruptions.

2.3.2 Usability

Usability requirements define how easy and efficient the system is for users to learn and operate.

- **USE-1:** The system shall allow a user to **create or access a case within three interactions** from the dashboard.
- **USE-2:** The AI Legal Chat Assistant shall provide responses in **clear natural language** understandable by users without legal expertise.
- **USE-3:** The system interface shall be **web-based and responsive**, allowing access from desktops, tablets, and mobile devices.
- **USE-4:** New users shall be able to learn basic system operations (case creation, document upload, and AI query) within **10 minutes of first use**.
- **USE-5:** The system shall display **clear error messages and guidance** whenever a user performs an invalid operation.

2.3.3 Performance

Performance requirements define how fast the system responds and handles workload.

- **PER-1:** The system shall respond to user requests (e.g., opening case records) within **2 seconds under normal load conditions**.
- **PER-2:** The AI Legal Chat Assistant shall generate responses within **5 seconds for standard legal queries**.
- **PER-3:** The system shall support **at least 100 concurrent users** without significant performance degradation.
- **PER-4:** Document uploads of up to **20 MB** shall be completed within **10 seconds on a standard broadband connection**.
- **PER-5:** Database queries related to case retrieval shall be executed within **2 seconds for 95% of requests**.

2.3.4 Security

Security requirements ensure protection of sensitive legal information and system access.

- **SEC-1:** The system shall enforce **secure user authentication** before allowing access to any system resources.
- **SEC-2:** Only authorized users shall be allowed to **view, edit, or delete case information** based on their assigned roles.
- **SEC-3:** All sensitive data such as **user credentials and case documents** shall be stored using secure encryption methods.

- **SEC-4:** The system shall log all major activities including **case updates, document uploads, and AI queries** for auditing purposes.
- **SEC-5:** The system shall protect against common web vulnerabilities including **SQL injection, cross-site scripting (XSS), and unauthorized API access**.

The Domain Model illustrates the structural data requirements and core entities of the LegalEase AI ecosystem. Following a professional modeling style, it uses inheritance to differentiate specialized user roles (Lawyer, Client, Student) from a base User entity. The diagram specifies detailed attributes (e.g., `ObjectId`, `datetime`, `json`) and precise multiplicities to define the relationships between cases, participants, schedules, and document assets.



Chapter 3

System Overview

LegalEase AI is a web-based platform designed to assist lawyers, clients, and law students by providing tools for managing legal information and accessing AI-based legal assistance. The system aims to simplify the process of organizing legal cases, storing documents, tracking deadlines, and improving communication between lawyers and clients. Instead of relying on manual methods to manage case records and documents, the platform provides a centralized digital environment where these tasks can be handled more efficiently.

The system mainly supports three types of users: **lawyers, clients, and students**. Lawyers can create and manage cases, upload documents, and monitor important deadlines. Clients can log into the system to view updates related to their cases and access shared documents. Law students can use the **practice simulator** to explore legal scenarios and improve their understanding of legal procedures.

In addition to case management, the platform includes an **AI Legal Chat Assistant** that allows users to ask legal questions and receive informative responses. The system also provides features such as document analysis, legal research, and notifications to support different legal activities. The platform is developed using modern technologies including **React.js for the frontend, FastAPI for backend services, and PostgreSQL for database management**. The system is also designed to support both **Urdu and English** to better handle legal documents commonly used in the local legal environment.

Overall, LegalEase AI aims to provide an organized and intelligent platform that helps users manage legal information more effectively while also improving access to legal knowledge.

3.1 Architectural Design

The LegalEase AI system is designed using a Layered + Microservices architectural pattern. This approach decomposes the application into specialized, self-contained modules that collaborate through well-defined interfaces. This modularity ensures that the system is scalable, maintainable, and aligned with the complex requirements of legal-tech intelligence.

3.1.1 Modular Program Structure

The system is decomposed into three primary layers, each with specific responsibilities:

- **Presentation Layer (Frontend):** Handles user interaction through role-specific dashboards (Lawyer, Client, Student). Implemented using React 19 to provide a dynamic, responsive, and professional user interface.
- **Application Layer (Backend Services):** Acts as the intelligence hub of the system. Uses an API Gateway pattern via FastAPI to manage routing, security (JWT), and communication between functional modules. Functional modules like the AI Document Analysis and Legal Research are isolated to allow independent scaling and specialized optimization.
- **Data & Storage Layer:** Manages persistence for structured legal data (Postgres), unstructured document files (S3/Local Storage), and AI-specific embeddings (Vector DB).

3.1.2 Module Collaboration & Relationships

The modules collaborate in a decoupled manner to achieve the system's core functionality:

- **Authentication Module:** Validates user identity and provides role-based access tokens to the entire system. **Case Management Module:** The central coordinator that links users to their legal cases, schedules hearings, and tracks documents.
- **AI Intelligence Modules:** These services (RAG Assistant, OCR, Summarizer, Simulator Engine) receive data from the core backend. They operate on a decoupled basis, allowing heavy AI processing to occur without affecting the responsiveness of the case management of UI.
- **Simulator Engine & Research:** Specialized modules for law students to practice in simulated environments and for lawyers to search against massive datasets of Pakistani statutes.

3.1.3 High-Level Box and Line Diagram:

The Box and Line diagram provides a full system overview, including the target AI services.

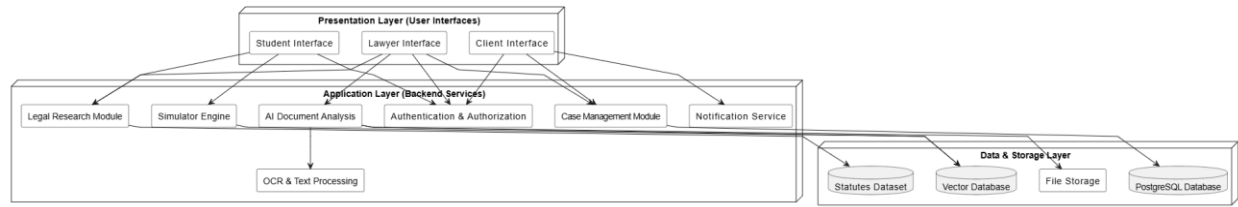


Figure: Box and Line Diagram

3.1.4 Layered + Micro Services Architecture Diagram:

This diagram illustrates the Layered + Microservices architectural pattern specifically designed for LegalEase AI. It organizes the system into four logical tiers: the User Interface Tier, the Gatekeeper Layer (API & Security), the Service Layer (decomposed into Core and AI-specific Microservices), and the Data Persistence Layer. This structure ensures that heavy AI computations are decoupled from the core case management logic, providing a scalable foundation for legal intelligence.

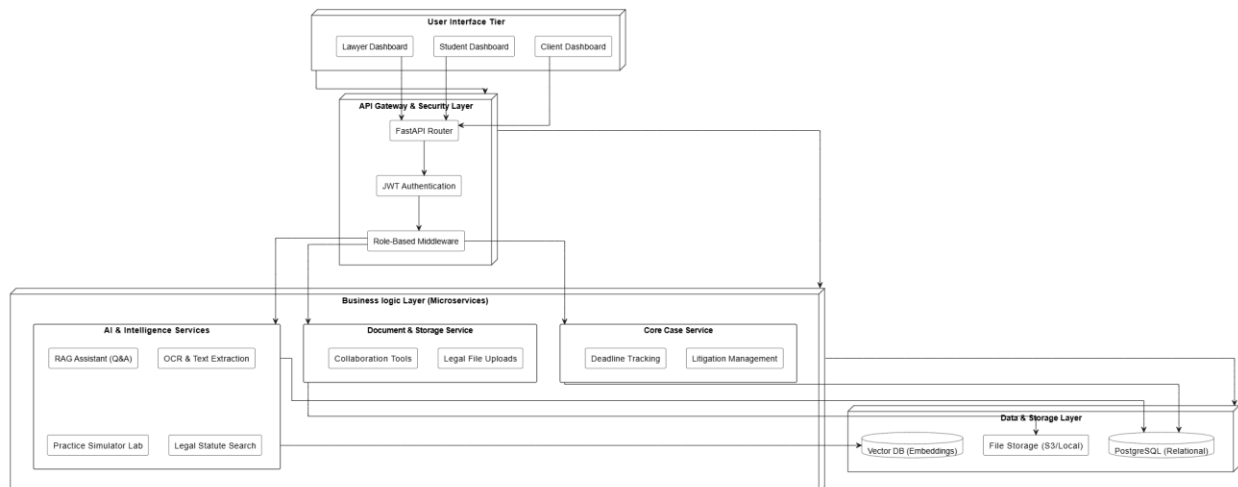


Figure: System Architecture Diagram

3.2 Design Models:

3.2.1 Activity Diagram

This section outlines the 10 core logical workflows of the LegalEase AI system:

Lawyer Uploads Case Document: Handles document ingestion with conditional OCR for scanned files and automated AI-driven clause and risk detection.

Client Upload Document: A streamlined entry point for clients to contribute files to a case, triggering immediate lawyer notifications. **AI Legal Research:** A semantic search pipeline for discovering Pakistani legal precedents and statutes using intelligent AI filtering.

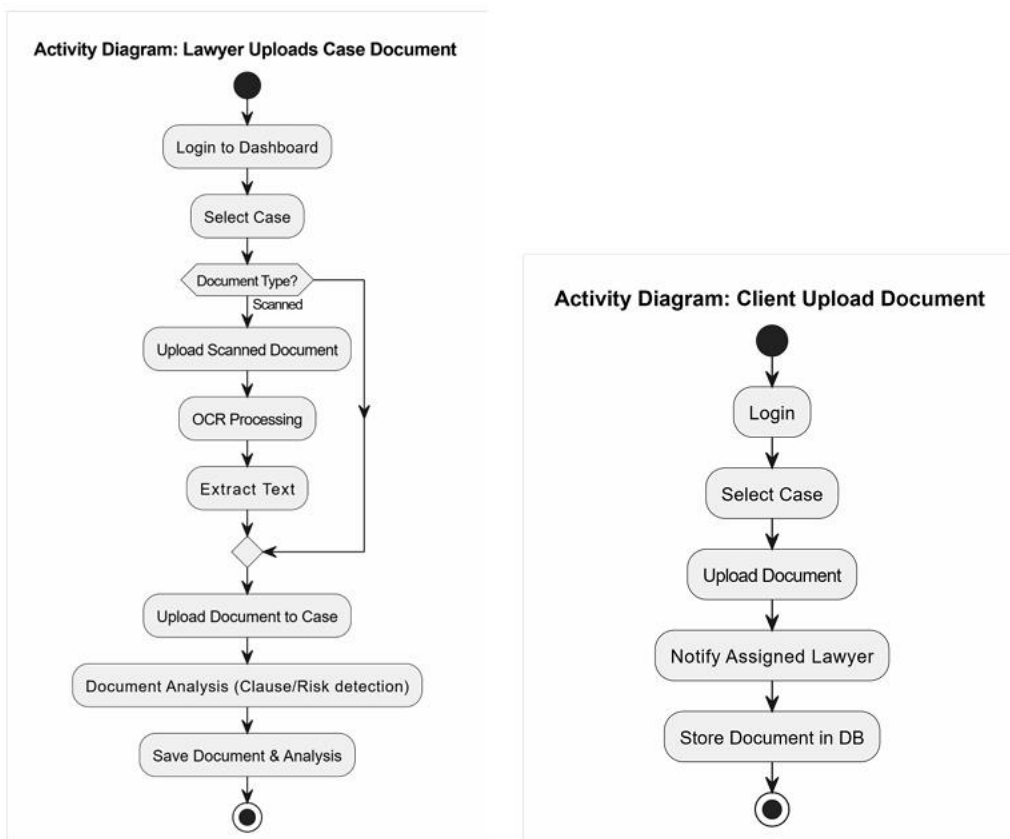


Figure: Lawyer Uploads Case Document and Client Upload Document

Contract Drafting & Review: A collaborative drafting cycle where AI suggests clauses, checks compliance, and manages version reviews.

AI Legal Chat Assistant: Real-time RAG-based Q&A flow that retrieves case context to provide accurate legal assistance.

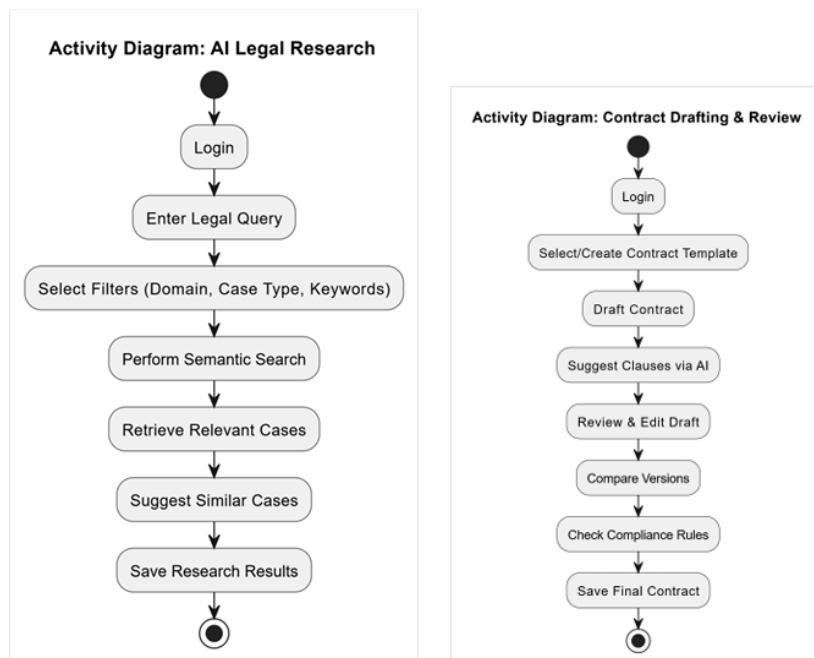


Figure:Contract Drafting & Review and AI Legal Chat Assistant

Practice Simulator: An interactive educational loop for students to practice legal scenarios and receive automated performance feedback.

AI Legal Chat Assistant: Real-time RAG-based Q&A flow that retrieves case context to provide accurate legal assistance.

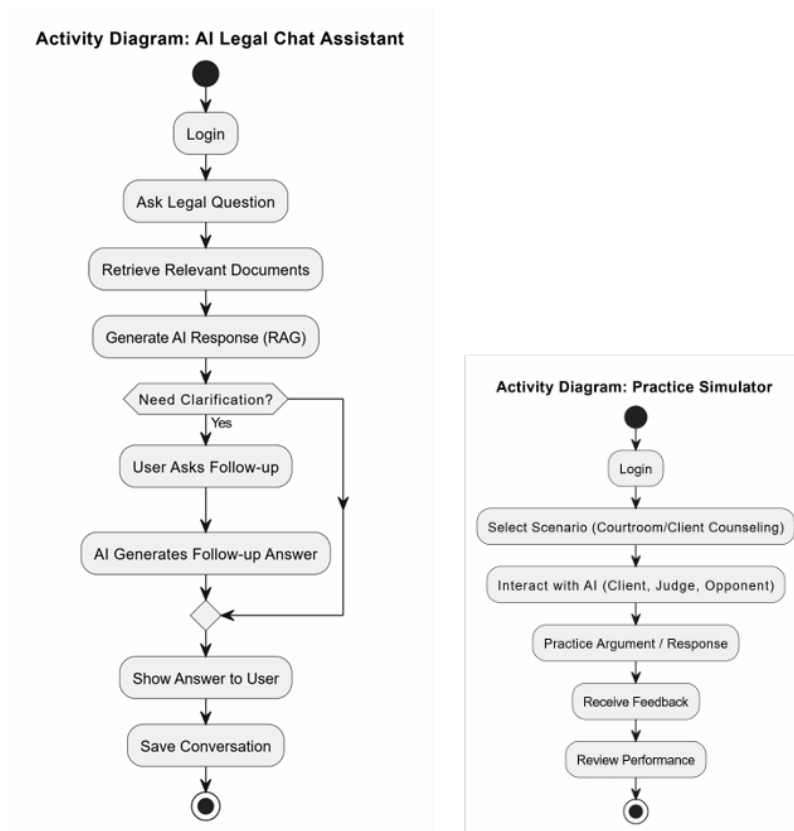


Figure: Practice Simulator and AI Legal Chat Assistant

OCR Processing: A technical service flow focused on the conversion and classification of physical legal documents into digital intelligence.

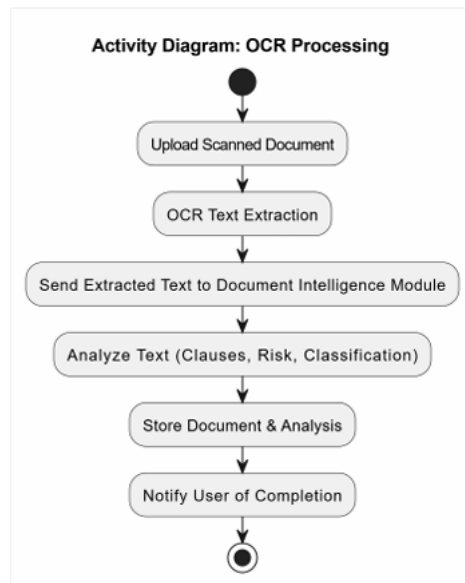


Figure: OCR Processing

Case Management - Create Case: The administrative setup for new litigation, including stakeholder assignment and deadline initialization.

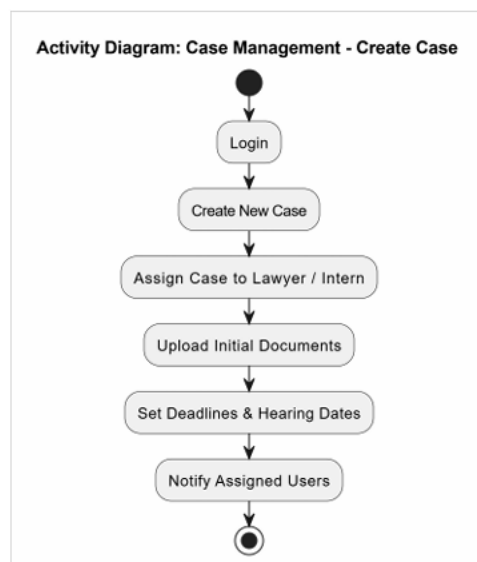


Figure: Case Management - Create Case

Track Case Progress: A monitoring workflow for viewing interactive milestones, hearing timelines, and automated reminders.

Admin Role Management: The system's control-plane workflow for managing user permissions, roles, and security audit logs.

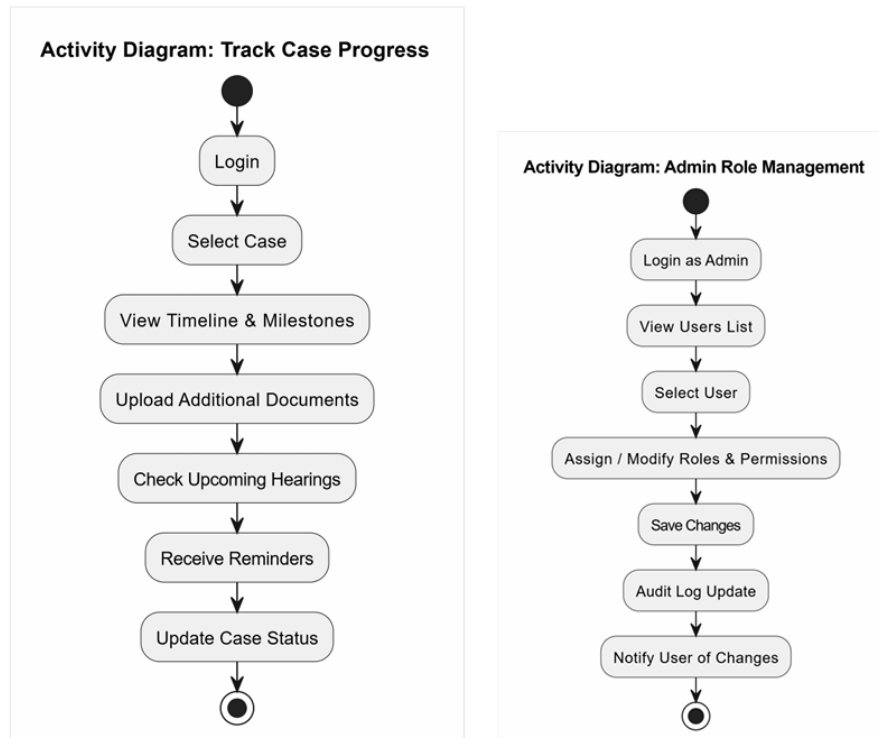


Figure:Track Case Progress and Admin Role Management:

3.2.2 Class Diagram

The Class Diagram represents the complete technical blueprint of the LegalEase AI system. It details the static structure of the application, including core business entities (Users, Cases, Documents), AI-driven components (Chat, OCR, Summaries), and specialized modules for Legal Education and Research. This diagram specifies all primary attributes and the complex relational associations (inheritance, composition, and aggregation) required to manage multi-role workflows and legal data persistence.

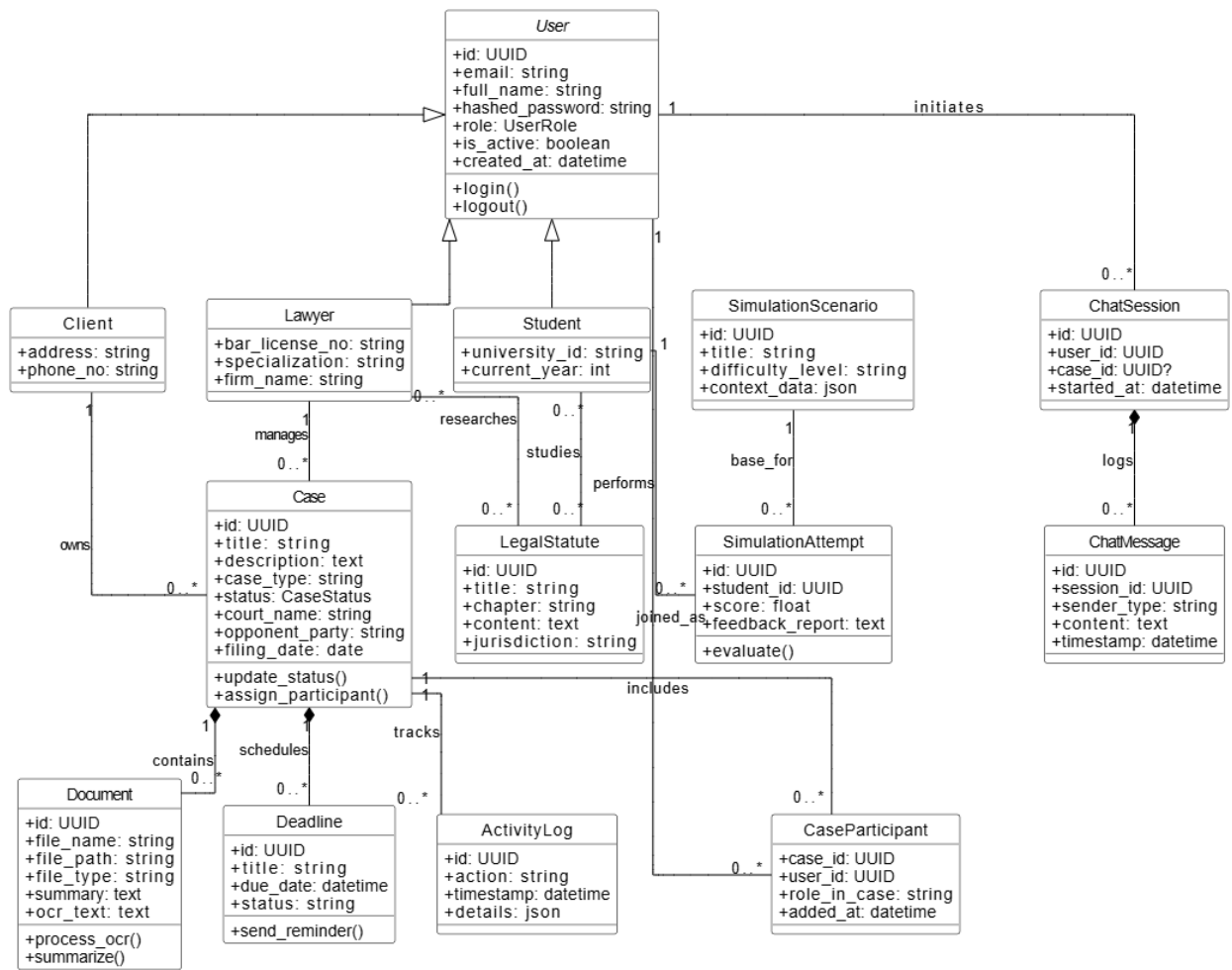


Figure: Class Diagram

3.2.3 Sequence Diagrams

The sequence diagrams show the step-by-step interaction between users and system components during the execution of system functionalities.

Load Role- Based Dashboard:

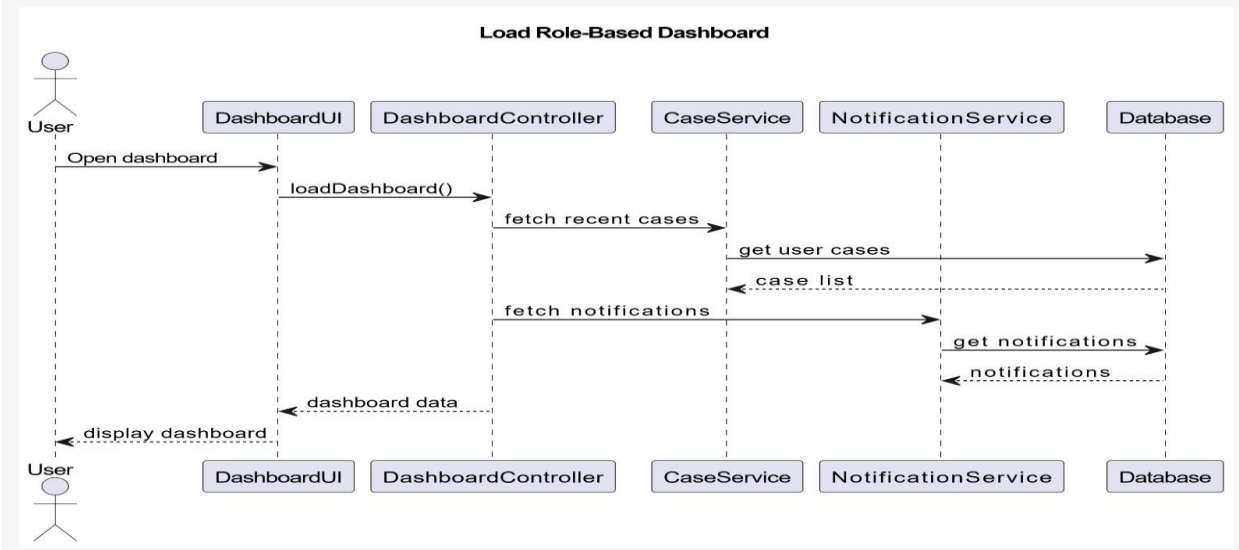


Figure: Load Role- Based Dashboard SD

User Login:

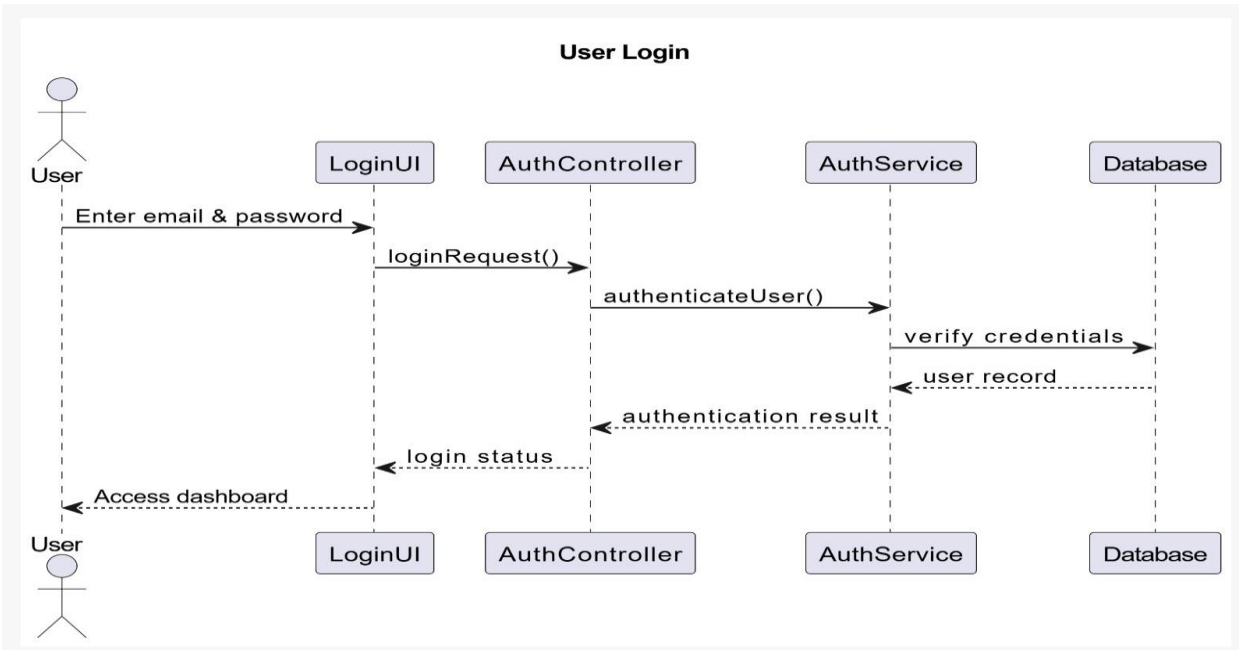


Figure: User Login SD

View Document Analysis Result:

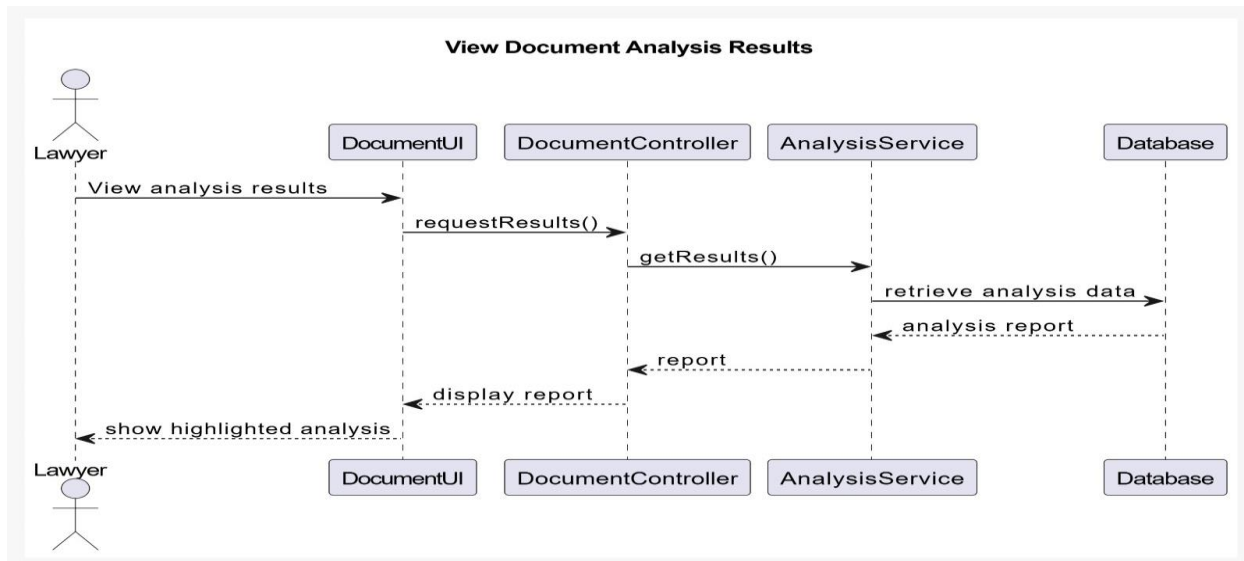


Figure:View Document Analysis Result SD

Assign Cases to Client:

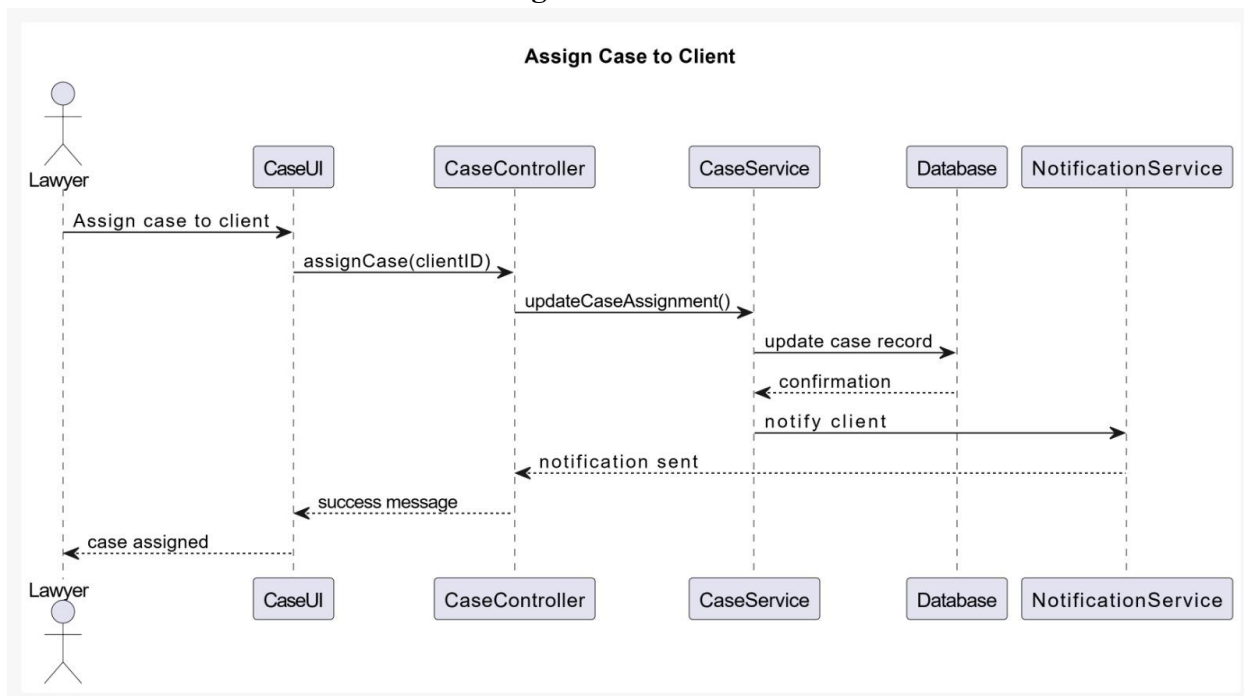


Figure: Assign Cases to Client

AI Document Summarization:

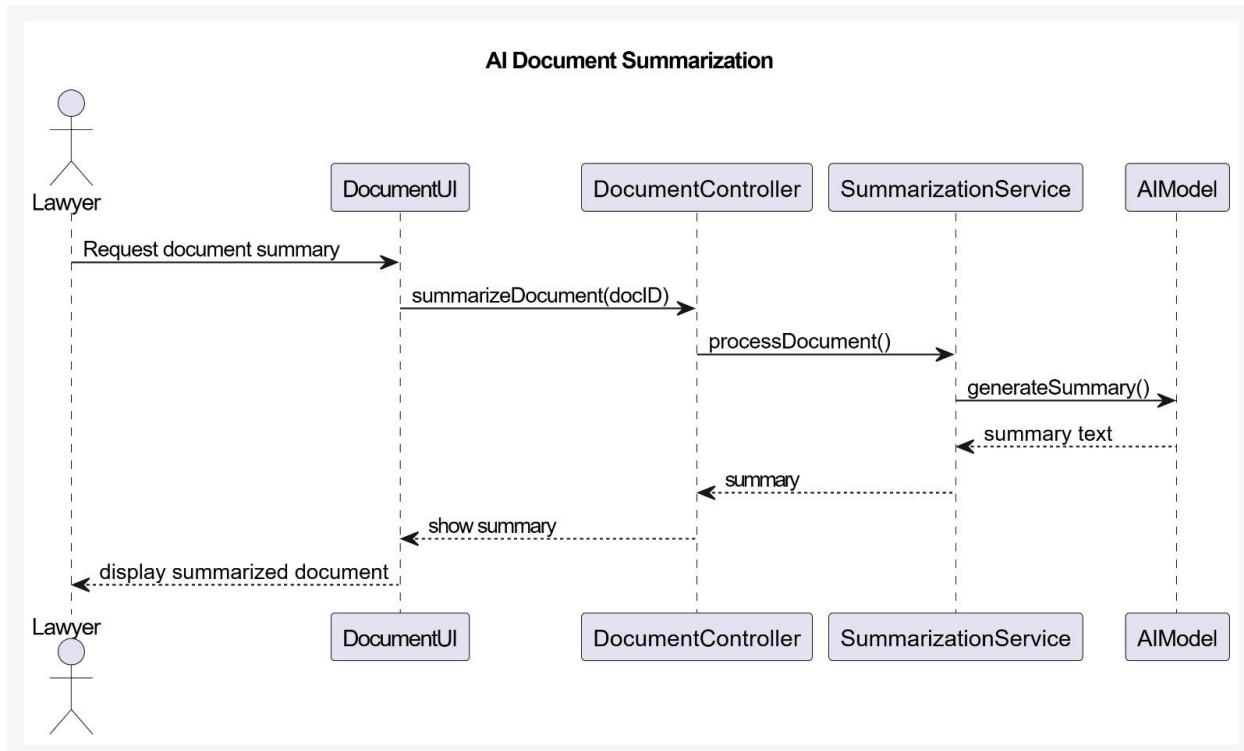


Figure: AI Document Summarization SD

Search Documents in Case:

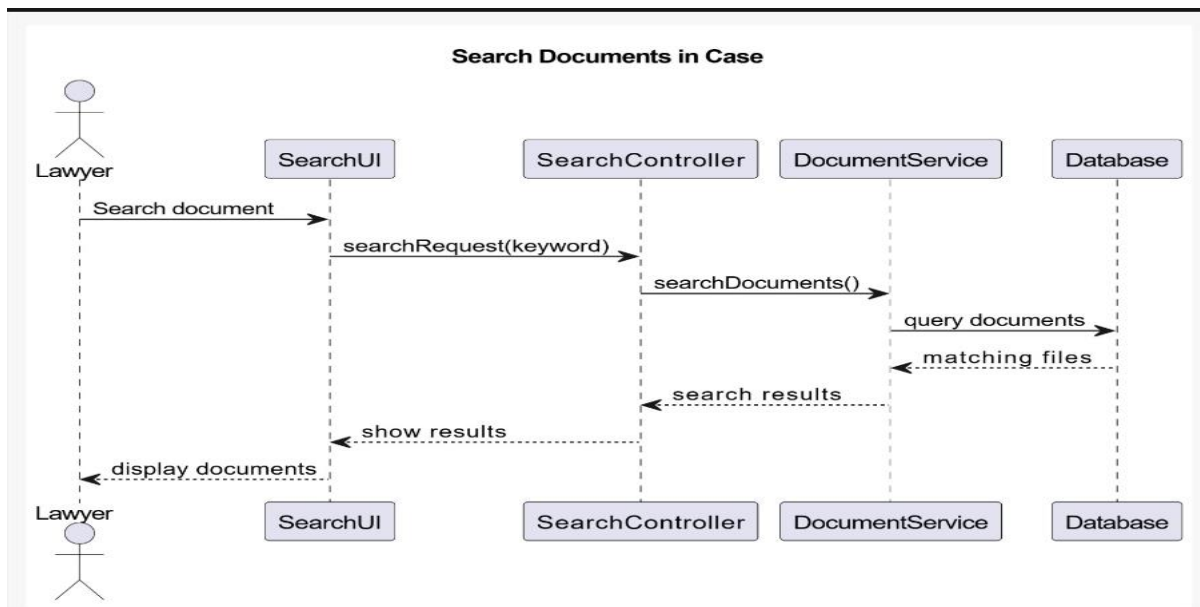


Figure: Search Documents in Case SD

Browsing:

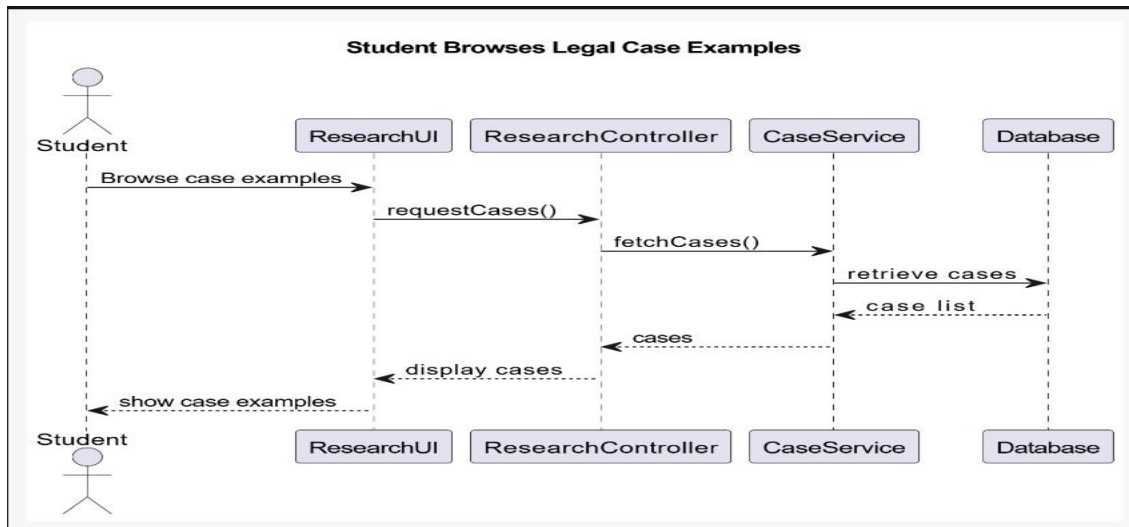


Figure: Browsing SD

Update Case Status:

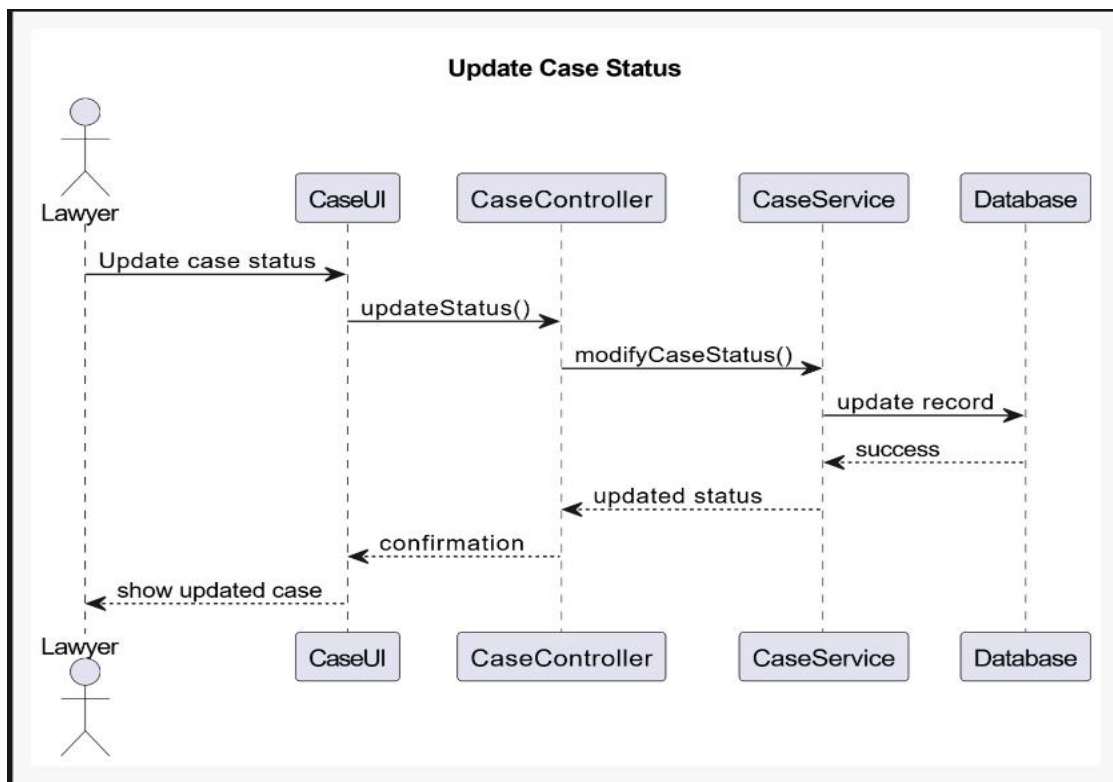


Figure: Update Case Status SD

View Notifications:

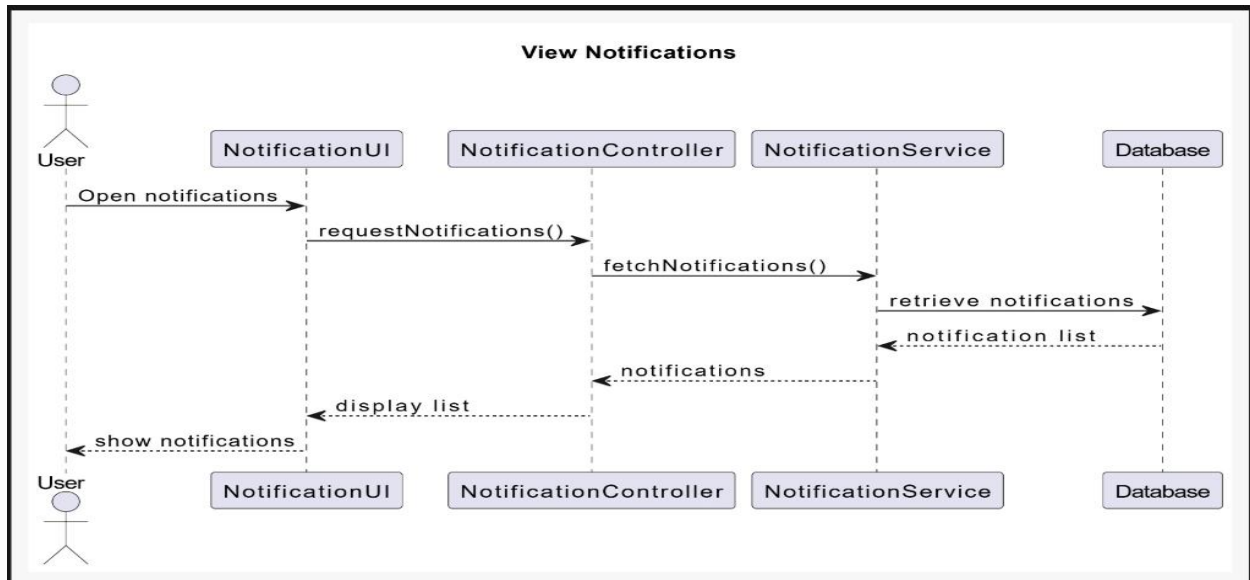


Figure: View Notifications SD

3.2.4 State Transition Diagram

Case Lifecycle: Tracks a legal case from its initial Created state through Assigned and InProgress phases, including the iterative HearingScheduled loop, until final Closed status.

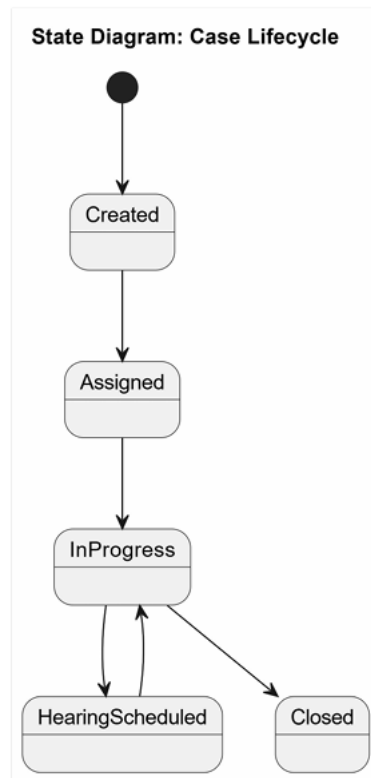


Figure:Case Lifecycle State Transition Diagram

Document Processing: A linear backend state machine for file intelligence, moving from Uploaded to OCRProcessing, TextExtracted, Analyzed, and finally Stored.

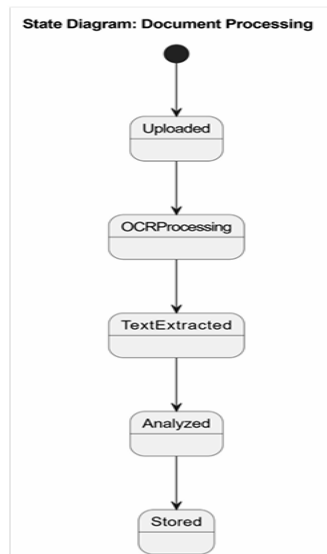


Figure: Document Processing Diagram

Contract Drafting: Manages the iterative refinement of legal documents, transitioning from Draft through UnderReview, Revised, and Approved, to the final Finalized state.

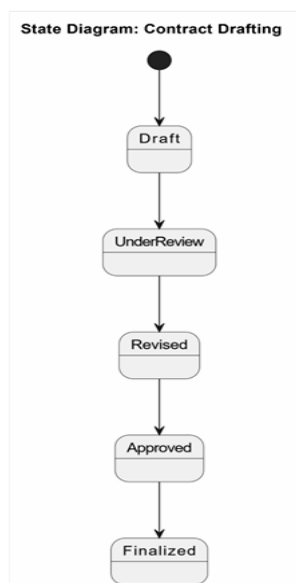


Figure: Contract Drafting State Transition Diagram

User Session: Controls the authentication state of the application, managing transitions between LoggedOut, LoggingIn, LoggedIn, and LoggingOut.

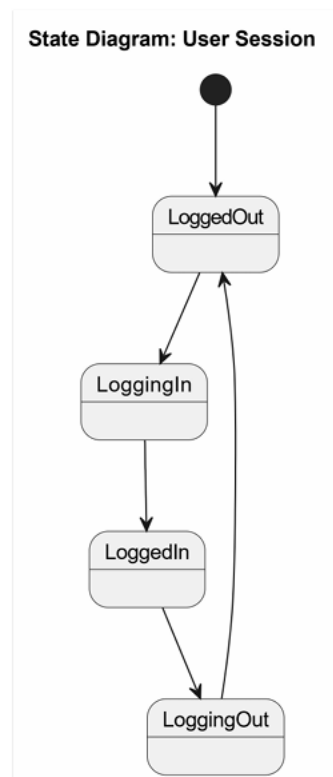


Figure: User Sesssion State Transition Diagram

Practice Session: The state flow for student educational labs, covering ScenarioSelected, SimulationRunning, ArgumentSubmitted, Evaluation, and final FeedbackGiven.

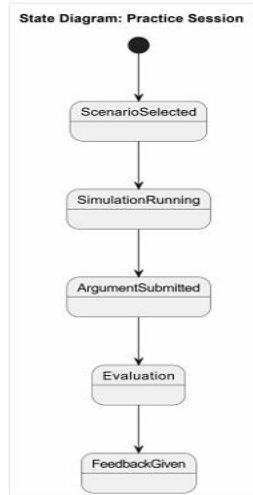


Figure: Practice Simulator State Transition Diagram

Notification: A reliable delivery state machine for system alerts, tracking them from Created through Sent and Delivered to the final Read state by the user.

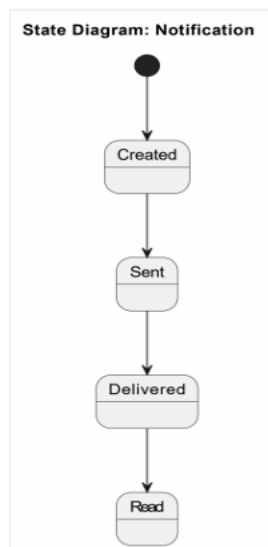


Figure: Notification State Transition Diagram

AI Chat Assistant: Models the real-time RAG intelligence loop, transitioning from QuestionAsked to RetrievingDocuments, GeneratingAnswer, AnswerDisplayed, and WaitingNextQuery.

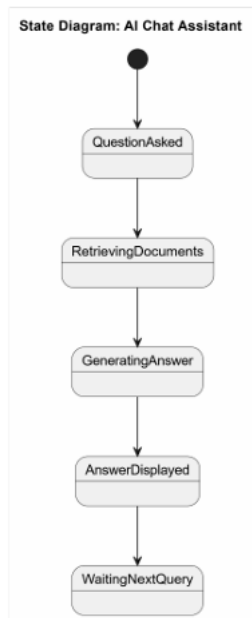


Figure: AI Chat Assistant State Transition Diagram

3.3 Data Design

The data design explains how the system organizes and manages information. In this project, user inputs and system data are stored in a structured database so that they can be easily accessed and updated when needed. The main data entities include **users, lawyers, students, legal cases, documents, practice simulations, and feedback**. These entities relate to different modules of the system. The **Practice Simulator module** stores simulation data for both lawyers and students so they can practice legal scenarios. All data is processed and stored in the database to support the different modules of the system and ensure smooth system operation.

3.3.1 Database Stack Information

- **PostgreSQL:** Our main relational database (as seen in the ERD). It handles users, cases, and schedules with strict data integrity.
- **Vector Database** (e.g., Pinecone or FAISS): Used specifically for the AI features. It stores the "embeddings" of legal documents so the AI can search them by meaning.
- **Blob Storage** (S3/FS): Used for the actual PDF and image files of cases.

3.3.2 Data Security

To protect sensitive legal information, the system implements:

- Encrypted password storage
- Role-based access control
- Audit logging for user activities
- Secure document storage and access permissions These measures ensure confidentiality, integrity, and controlled access to legal data

ERD:

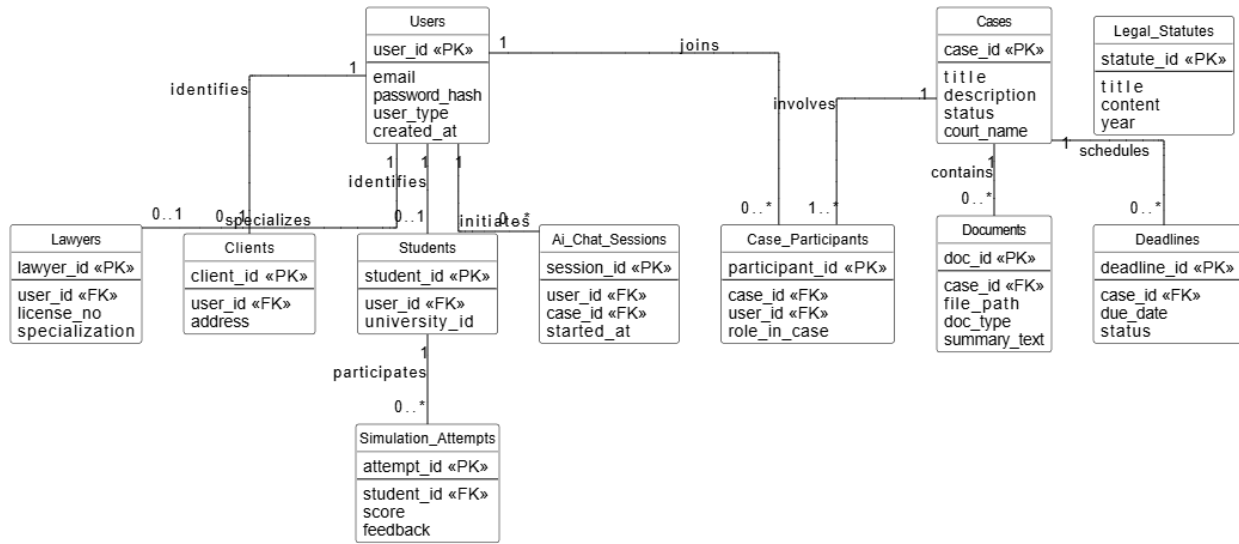


Figure: ERD

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